Mab characterization

- Concentration of pure antibody

- The concentration of pure antibodies can be estimated from the measured absorbance at 280nm, assuming a value of 13 to 14 for the absorbance of a 1% (10 mg/mL) solution in saline or PBS.

- The concentration of pure antibodies also can be estimated using any commercial protein assay, such as our BCA Protein Assay Kit or Coomassie Plus Protein Assay Kit.

- Molecular weight determination: by SDS-PAGE or gel filtration chromatography
- Purity: Electrophoretic mobility
- Antibody consist of 2L and 2 H chains
- separate chains from each other, and run on SDS page 2 bands appear
- Pure Antibody: two bands appear when run on SDS page
- SDS-PAGE gel and gel filtration analysis showed that the antibodies from the hollow fiber bioreactors. Are very clean (pure / no contamination)

Immunological Properties for MAB test by :

- Cross reaction
- Titer
- Affinity
- Isotype
- Specificity

1. **Cross reactivity**

- To test whether the separated clones are identical that produce identical Mabs, and reacts with the same epitopes, or totally different Mabs that react with different epitopes, an inhibition radioimmunoassay is performed.

- Coat all wells with constant amount of the Ag.

- Add serial dilutions from zero to saturating levels of the specific unlabelled Mab1 (add before Labeled Ab)

- Add constant amount of iodine-labelled Mab2-I125 and incubate to allow it to compete for the Ag

- Wash the excess labelled Mab2.

- If the labelled Mab2 does not bind the Ag and washed out, the two Mabs recognize the same epitopes and are identical.

- If the labelled Mab2 bind the Ag and does not appear in the wash, the 2 Mabs are different.

- If patial inhibition occurs, both recognize close epitopes

- Results should be interpreted carefully, since a high affinity labelled Mab may displaced a low affinity Mab at the same epitope

- In simple word, we coat the wells with Antigens, then add serial dilution of un labeled Ab in the wells, then wash, then add constant a mount of Labeled Ab (labeled by iodine 125), then wash, suppose the Ag contain Two epitopes, if the labeled Ab found in Washing this mean both (labeled and un labeled Ab recognize the same epitopes)

- If the labeled Ab not present in Washing, Labeled And Un labeled Ab recognize each recognize different epitope for the same Ag.

2. **Titer determination of the Mab titer**

- The titer of an Ab is a measure of its concentration under defined conditions

- The titer is defined as the lowest dilution of Mab that bind significantly with an Ag in ELISA or

- The dilution of Mab that gives half maximal activity (binding to Ag).

- It involves serial dilutions of the Mab are allowed to react with known constant amount of an Ag

Procedure

1-Constant amount of Ag coating of microtiter plate wells

2-Washing of unbound Ag

3-Add the Mab in serial dilutions and Wash

4-Add Enzyme-Ab conjugate and wash

5-Add substrate

6-Read the absorbance

Determination of Mab Specificity

- A robust ELISA would benefit when the Abs do not crossreact with non-target molecules, closely related metabolic products and homologous interfering molecules.

- For example, Human serum albumin ELISA kit is specific for HAS and does not react with mouse, rat, and rabbit albumin.

- **Principle**:

- Coating the microtiter plate wells with different Ags related to the target Ag

- Performing ELISA

Isotype determination

Principle of the Assay:

- Using Monoclonal Antibodies against constant regions

- Isotyping involves determining the class (e.g., IgG vs. IgM) and subclass (e.g., IgG1 vs. IgG2a) of a Mab.

- Isotyping is most easily accomplished with commercial, ready-to-use Ab isotyping kits.

- The method employs the quantitative sandwich enzyme immunoassay technique.

- Capture anti-isotype antibodies for mouse: heavy-chain class-specific Abs (anti-μ, -α, -γ, -δ, -ε), heavy-chain subclassspecific antibodies (anti-γ1, -γ2a,-γ2b, -γ3), or light-chain isotype-specific Abs (anti-κ, -λ)

- Samples are pipetted into microwells and Ig present in the sample is bound by the capture antibody.

- Then, a HRP (horseradish peroxidase) conjugated antimouse IgG (H+L) antibody is pipetted and incubated simultaneously with samples.

- After washing microwells in order to remove any nonspecific binding, the ready to use substrate solution (TMB) is added to microwells and color develops if the specific immunoglobulin is present in the sample.

- Color development is then stopped by addition of stop solution. Absorbance is measured at 450 nm.

Affinity determination

- Affinity is the description of the strength of interaction between a Mab and an Ag.

- Interaction of high affinity (K value of 10^9-10^12) are essential for use in immunoassays where the bond between Ag and the Mab has to withstand vigorous washing procedures.

- For immuno purification, low affinity Mab (K value 106-107) are preferred where the bond between Ag and Ab can be easily broken when necessary.

- Ag + Mab · Ag-Mab complex

Affinity constant K = Ag-Mab complex

----------------------------free Ag x free Mab

- Data obtained using labelled Ag should be interpreted with caution because labelling process may destroy the epitope to which the Mab is directed.

- A rough estimate of affinity can be deduced by comparing Ig conc with the strength of the measured response. E.g if there is strong specific Ab response, the Ag-Ab reaction is likely to be of higher affinity

Inhibition RIA

- Serial dilutions of unlabeled Ag are dispensed into assay tubes
- Constant labelled Ag is added to each
- Constant Mab is added to all wells
- After incubation, the second Ab is added to all
- Centrifuge and separate the supernatant.

- Quantify the radioactivity in the pellet by gamma counter.

- The free Ag, free Mab and the Ag-Mab complex can be determined.

- Scatchard plot: Plot the ratio of bound Ag to free Ag against bound Ag. A straight line of negative slope should be obtained.

- If a straight line is not obtained, the Mab may not be monoclonal

K= 1/slope or y/x

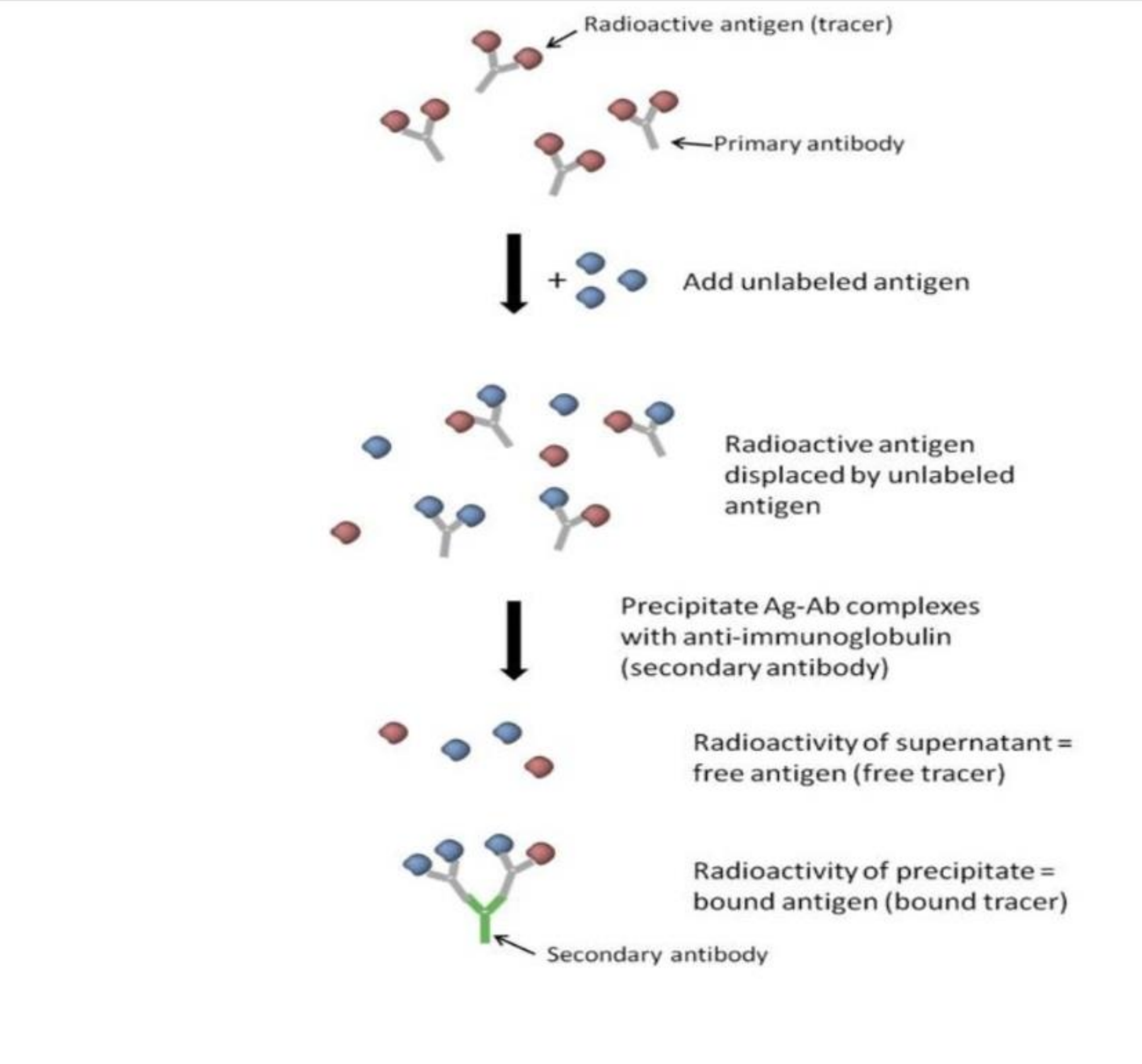
Measurement of Mab affinity

1.The ELISA plate was coated with 2mg/mL of OVA. After incubation for a 15h period at 4C, the plate was washed three times, then blocked by adding dilution buffer, and incubated for 1h at room temperature.

2.At the same time, OVA antigen at various concentrations (from3·10–9M to1·10–7M) was incubated with anti-OVA antibodies at constant concentration (deduced from preliminary ELISA calibration for each Mab) for 15h at room temperature and100mL of each mixture was transferred into the wells of the ELISA plate previously coated with OVA and incubated for 1h at 370C.

3.After washing, 1:2500 diluted HRP-conjugated goat anti-mouse IgG was added and the plate was incubated for 45 min at 37C.

4.Finally, the plate was washed five times and then observed with TMB. The dissociation constant (KD) was calculated according to the following formula.


Radioactive antigen (tracer)
Primary antibody
+
Add unlabeled antigen
Radioactive antigen
displaced by unlabeled
antigen
Precipitate Ag-Ab complexes
with anti-immunoglobulin
(secondary antibody)
Radioactivity of supernatant =
free antigen (free tracer)
Radioactivity of precipitate =
bound antigen (bound tracer)
Secondary antibody

Tube	No.	CPM	Avg CPM	Net Avg CPM	%B/B0
Total Counts	1	14963			
	2	15563	15263		
Blank	3	415			
	4	380	398		
"0" Standard	5	8879			
	6	8673	8776	8378	100
2 pg	7	7701			
	8	7939	7820	7422	88.6
5 pg	9	6873			
	10	6886	6880	6482	77.4
10 pg	11	5632			
	12	5900	5766	5368	64.1
25 pg	13	4216			
	14	4256	4236	3838	45.8
50 pg	15	3082			
	16	2995	3039	2641	31.5
100 pg	17	2257			
	18	2003	2130	1732	20.7

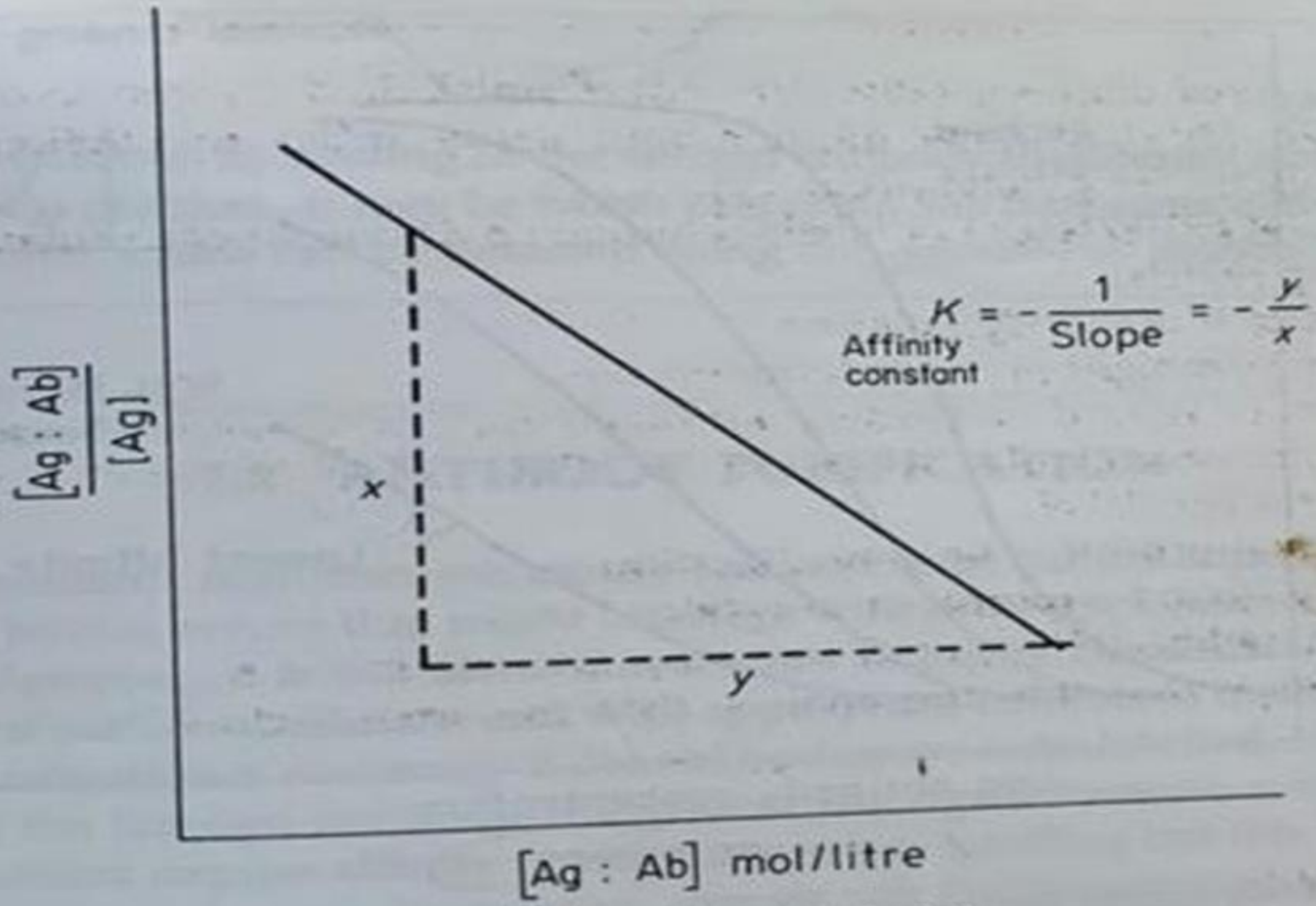


Fig. 7.1.4a Scatchard plot